

Comprehensive Topography of YouTube and Google Analytics API Systems: A Metadata and Metrics Reference Architecture

The modern digital video ecosystem operates on a foundation of hyper-granular data telemetry. The YouTube data infrastructure is not a singular entity; it is a multi-tiered architecture comprising the YouTube Data API v3, the YouTube Analytics API, and the YouTube Reporting API, further augmented by external integrations like Google Analytics 4 (GA4).

This report provides an exhaustive, re-formatted reference architecture of every available data point, metric, and dimension. The data is logically organized into architectural dimensions, universal metrics, format-specific analytics (Shorts vs. Long-form), metrics restricted to Studio UI/CSV exports, a guide for external traffic routing, and a comprehensive dictionary of advanced derived statistical formulas.

1. Systemic Architecture of YouTube Data Interfaces

To parse the metrics accurately, the structural distinctness of the systems delivering them must be established:

- **YouTube Data API v3:** Manages channel structural hierarchy and metadata (titles, descriptions, publishing status). It provides basic, public-facing aggregate statistics (like total channel views or subscriber counts) but is not designed for deep behavioral analysis.
- **YouTube Analytics API:** Designed for real-time, targeted queries, allowing systems to generate custom reports. It utilizes camelCase formatting for its data objects (e.g., `averageViewDuration`, `trafficSourceType`). It is heavily constrained by query quotas.
- **YouTube Reporting API:** Built for asynchronous, bulk data extraction. It generates daily, system-managed CSV-style reports where each row's unique combination of dimensions acts as a primary key. It utilizes snake_case formatting (e.g., `average_view_duration_seconds`, `traffic_source_type`).

Across the analytics APIs, data fields are categorized as **Dimensions** (the criteria used to aggregate and group data) and **Metrics** (the quantitative measurements). "Core" dimensions and metrics are fundamental to the API and are protected by Google's Deprecation Policy, ensuring they will not be removed without substantial advance notice.

2. Core and Additional Dimensions: The Axes of

Measurement

Metrics exist in a vacuum until intersected with Dimensions. Dimensions aggregate the metrics across time, geography, user demographics, and hardware.

Temporal and System Dimensions

Common Name	Analytics API Title	Reporting API Title	Description & Utility
Day	day	date	Groups data by calendar day (12:00 AM to 11:59 PM Pacific Time).
Month	month	N/A	Aggregates data by calendar month (Analytics API only).
Video ID	video	video_id	The unique 11-character alphanumeric identifier for a specific video.
Channel ID	channel	channel_id	The unique identifier for a channel, frequently used in Content Owner reports.
Playlist ID	playlist	playlist_id	Isolates metrics generated within the sequential playback of a curated list.
Content Type	creatorContentType	N/A	Identifies media format: LIVE_STREAM,

			SHORTS, STORY, VIDEO_ON_DEMAND. ¹
Uploader Type	uploaderType	uploader_type	Distinguishes between owner uploads (self) and third-party uploads (thirdParty).
Asset ID	N/A	asset_id	Represents intellectual property (e.g., a song) within Content ID.

Audience Demographic Dimensions

Common Name	Analytics API Title	Reporting API Title	Description & Utility
Age Group	ageGroup	age_group	Standardized age brackets (including under-18 estimations as of 2026). ²
Gender	gender	gender	Inferred or self-identified gender of logged-in users.
Subscription Status	subscribedStatus	subscribed_status	Indicates if the user was SUBSCRIBED or UNSUBSCRIBED precisely during the view.

Geographic Dimensions

Common Name	Analytics API Title	Reporting API Title	Description & Utility
Country	country	country_code	Two-letter ISO-3166-1 country code (e.g., US, GB).
Province / State	province	province_code	Identifies specific U.S. states using ISO 3166-2 codes.

Playback, Traffic, and Hardware Dimensions

Common Name	Analytics API Title	Reporting API Title	Description & Utility
Playback Location Type	insightPlaybackLocationType	playback_location_type	Categorizes the UI where the video played (e.g., WATCH, EMBEDDED, SHORTS).
Playback Location Detail	insightPlaybackLocationDetail	playback_location_detail	Provides the specific URL or app for EMBEDDED playbacks.
Traffic Source Type	insightTrafficSourceType	traffic_source_type	Defines the referrer intent (e.g., YT_SEARCH, RELATED_VIDEO, EXT_URL).

Traffic Source Detail	insightTrafficSourceDetail	traffic_source_detail	Provides granular referrer details (e.g., exact search query, exact external URL).
Device Type	deviceType	device_type	Categorizes hardware: Computer, TV, Game Console, Mobile Phone, Tablet.
Operating System	operatingSystem	operating_system	Software architecture (e.g., Windows, iOS, Android).
Sharing Service	sharingService	sharing_service	Identifies the external platform used via the native "Share" button.

3. Universal Base Metrics

These quantitative measurements apply to all content across the platform.

View, Reach, and Engagement Metrics

Common Name	Analytics API Title	Reporting API Title	Description
Total Views	views	views	The raw count of video views. For Shorts, this counts the moment playback begins. ²
Engaged Views	engagedViews	engaged_views	Views that bypass initial scrolling and

			persist past the first few seconds. ²
Premium Views	redViews	red_views	Views generated exclusively by authenticated YouTube Premium members.
Logged-in View %	viewerPercentage	views_percentage	Percentage of viewers actively logged into a Google account during the view.
Likes	likes	likes	Total positive user ratings.
Dislikes	dislikes	dislikes	Total negative user ratings (tracked internally despite consumer UI hiding).
Comments	comments	comments	Total absolute number of comments.
Shares	shares	shares	Number of times a video was distributed via the native "Share" button.
Subscribers Gained	subscribersGained	subscribers_gained	Users who subscribed during the queried period.

Watch Time and Revenue Metrics

Common Name	Analytics API Title	Reporting API Title	Description
Total Watch Time	estimatedMinutesWatched	watch_time_minutes	Aggregate duration of content consumed, measured in minutes.
Average View Duration	averageViewDuration	average_view_duration_seconds	Average length of a video playback, expressed in seconds.
Avg. Percentage Viewed	averageViewPercentage	average_view_duration_percentage	Average percentage of the video's total length watched during a playback.
Estimated Revenue	estimatedRevenue	estimated_partner_revenue	Total estimated net revenue from all sources (Ads + Premium + Transactions).
Ad Revenue	estimatedAdRevenue	estimated_partner_ad_revenue	Net revenue derived specifically from Google-sold advertising.
Transaction Revenue	N/A	estimated_partner_transaction_revenue	Revenue from Fan Funding, Super Chats, and Memberships.

Ad Impressions	adImpressions	ad_impressions	Verified number of times an advertisement was served and viewed.
Gross CPM	cpm	estimated_cpm	Estimated gross revenue generated per 1,000 ad impressions.

4. Format-Specific Base Metrics

Due to vastly different user interfaces, Long-form videos and Shorts generate exclusive metrics.

Long-Form Only Base Metrics

Long-form focus heavily relies on sustained retention, interactive on-screen overlays, and playlist continuity.

Common Name	Analytics API Title	Reporting API Title	Description
Card Impressions	cardImpressions	card_impressions	Times interactive mid-video info-cards were rendered on screen.
Card Clicks	cardClicks	card_clicks	Total clicks registered on displayed cards.
End Screen Impressions	N/A	end_screen_element_impressions	Times linked videos/subscribe buttons were shown in the final 20 seconds.

End Screen Clicks	N/A	end_screen_element_clicks	Total clicks on end screen elements.
Playlist Starts	playlistStarts	playlist_starts	Times a user initiated a sequential playlist playback session.
Playlist Saves	playlistSaves	playlist_saves_added	Times users saved a playlist to their personal library.

Short-Form Only Base Metrics

Shorts eliminate the "click" in favor of the "swipe," demanding metrics based on feed rendering and instantaneous hook rates.

Common Name	Analytics API Title	Reporting API Title	Description
Shown in Feed	videoThumbnailImpressions	video_thumbnail_impressions	Times a Short was physically rendered and presented in the Shorts feed. ²
Remix Traffic	insightTrafficSourceType	traffic_source_type	Identified when traffic source equals VIDEO_REMIXES, tracking views originating from secondary videos sampling the original audio. ¹
Sound Page Traffic	insightPlaybackLocationType	playback_location_type	Identified when playback location equals SOUND_PAGE. ³

5. Metrics Exclusive to Studio UI & Bulk CSV Reports

Certain critical community and acquisition metrics are natively restricted from the real-time Analytics API endpoints and can only be sourced via YouTube Studio dashboard exports or specialized bulk data warehousing.

Common Name	Description & Availability
New Viewers	Viewers who watched your channel for the very first time during the selected time period. Crucial for measuring top-of-funnel reach.
Returning / Regular Viewers	Viewers who have watched content on your channel previously and returned during the selected period. Measures core audience loyalty.
Casual Viewers	Viewers who watch the channel occasionally (at least once a month for 1-5 months in the past year).
Unique Viewers	The exact count of individual, authenticated human viewers (de-duplicated from raw view counts). This forms your "Monthly active audience".
Viewed vs. Swiped Away (%)	The pre-calculated percentage metric natively provided by YouTube detailing hook-success for Shorts. (Note: While developers must manually calculate this via the API using engagedViews and videoThumbnailImpressions, Studio provides it

	natively).
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6. Guide: Extracting and Displaying External Traffic Sources

To programmatically display external traffic drivers (e.g., rendering "reddit.com 2.5%" on a dashboard), you must use dimension intersections.

- Filter by Source Type:** Set the insightTrafficSourceType (Analytics API) or traffic_source_type (Reporting API) to EXT_URL.⁴
- Request Source Details:** Add insightTrafficSourceDetail (or traffic_source_detail) as your secondary dimension. For EXT_URL, this detail field automatically populates with the referring domain string (e.g., reddit.com, facebook.com).⁴
- Query Metrics:** Request the views metric. The API will return the raw view count paired with the domain.
- Calculate Share:** To get the percentage, you must make a secondary query for total video views without the EXT_URL filter, then apply the formula:

$$\text{Domain Share (\%)} = \left(\frac{\text{Domain Specific Views}}{\text{Total Video Views}} \right) \times 100$$

7. Advanced Derived Statistics & Formulas

Raw data often fails to provide strategic context. By utilizing the exact API parameters listed above, analysts can calculate these 20 advanced derived metrics to evaluate hook rates, algorithm dependency, and financial efficiency.

Audience & Retention Formulas

Statistic Title	Exact API Fields Required	Calculation Formula
Shorts Hook Rate	engagedViews, videoThumbnailImpressions	$\left(\frac{\text{engagedViews}}{\text{videoThumbnailImpressions}} \right)$

Shorts Swipe-Away Rate	engagedViews, videoThumbnailImpressions	$\left(1 - \frac{\text{engagedViews}}{\text{videoThumbnailImpressions}} \right)$
Avg. Views Per Viewer	views, Unique Viewers (CSV)	$\frac{\text{views}}{\text{Unique Viewers}}$
Returning Viewer Ratio	Returning Viewers, New Viewers (CSV)	$\left(\frac{\text{Returning Viewers}}{\text{New Viewers} + \text{Returning Viewers}} \right)$
Authentication Rate	viewerPercentage (Natively provided %)	Matches native viewerPercentage output.

Engagement & Interactivity Formulas

Statistic Title	Exact API Fields Required	Calculation Formula
Overall Engagement Rate	likes, comments, shares, views	$\left(\frac{\text{likes} + \text{comments} + \text{shares}}{\text{views}} \right)$
Dislike Ratio	dislikes, likes	$\left(\frac{\text{dislikes}}{\text{likes} + \text{dislikes}} \right)$
Comment-to-View Ratio	comments, views	$\left(\frac{\text{comments}}{\text{views}} \right)$
Share-to-View Ratio	shares, views	$\left(\frac{\text{shares}}{\text{views}} \right)$

End Screen CTR	end_screen_element_clicks, end_screen_element_impressions	$\left(\frac{\text{end_screen_element_clicks}}{\text{end_screen_element_impressions}} \right)$
Card CTR	cardClicks, cardImpressions	$\left(\frac{\text{cardClicks}}{\text{cardImpressions}} \right)$
Teaser Efficacy Rate	cardTeaserClicks, cardTeaserImpressions	$\left(\frac{\text{cardTeaserClicks}}{\text{cardTeaserImpressions}} \right)$
Annotation Close Rate	annotationCloses, annotationImpressions	$\left(\frac{\text{annotationCloses}}{\text{annotationImpressions}} \right)$

Algorithm & Traffic Context Formulas

Statistic Title	Exact API Fields Required	Calculation Formula
Search Dependency Ratio	views (Filtered by YT_SEARCH), views	$\left(\frac{\text{Views from YT_SEARCH}}{\text{views}} \right)$
Subscriber Conversion	subscribersGained, views	$\left(\frac{\text{subscribersGained}}{\text{views}} \right)$
Views Per Playlist Start	playlistViews, playlistStarts	$\frac{\text{playlistViews}}{\text{playlistStarts}}$
Peak-to-Average Live	peakConcurrentViewers, averageConcurrentViewers	$\frac{\text{peakConcurrentViewers}}{\text{averageConcurrentViewers}}$

Shorts Viral Index	views, likes, comments, shares, day	$\left(\frac{\text{views}}{\text{Days Since Published}} \right)$
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Monetization & Financial Formulas

Statistic Title	Exact API Fields Required	Calculation Formula
True Overall RPM	estimatedRevenue, views	$\left(\frac{\text{estimatedRevenue}}{\text{views}} \right)$
Ad-Only CPM	grossRevenue, adImpressions	$\left(\frac{\text{grossRevenue}}{\text{adImpressions}} \right)$
Geographic RPM	estimatedRevenue, views (Filtered by country)	$\left(\frac{\text{estimatedRevenue for Country}}{\text{views for Country}} \right)$
Monetized Penetration	monetizedPlaybacks, views	$\left(\frac{\text{monetizedPlaybacks}}{\text{views}} \right)$
Premium Revenue Share	estimatedRedPartnerRevenue, estimatedRevenue	$\left(\frac{\text{estimatedRedPartnerRevenue}}{\text{estimatedRevenue}} \right)$

8. External Analytics: GA4 and Ecosystem Integrations

YouTube's native APIs explicitly track telemetry *on* the platform. However, a holistic data strategy requires understanding how embedded video assets perform across the broader internet using Google Analytics 4 (GA4).

GA4 utilizes event-based tracking rather than session-based tracking, categorizing video

interactions into distinct events:

- video_start: Fired the moment the embedded video begins playing.
- video_progress: Fired at specific percentage milestones (e.g., 10%, 25%, 50%, 75%) to track external retention.
- video_complete: Fired when the video reaches its conclusion.

By creating custom metrics within GA4—such as assigning a binary 1 to a video_view parameter triggered specifically by the video_start event—analysts can quantify external engagement and cross-reference it with acquisition channels.

9. Strategic Synthesis

The true value of this exhaustive API architecture lies in the correlative synthesis of overlapping dimensions. Long-form video operates as a "Commitment Engine," relying on the friction of the click (videoThumbnailImpressionsClickRate) and the sustained psychological investment of watch time (averageViewDuration). Conversely, Short-form video operates as a "Density Engine," where the metric of entry is immediate algorithmic exposure (videoThumbnailImpressions) and hook retention (engagedViews). Mastery of these schemas allows for the precise, empirical reverse-engineering of user behavior and digital monetization.

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